

When to (or not to) trust intelligent machines: Insights from an evolutionary game theory analysis of trust in repeated games

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Abstract

The motives behind the actions of intelligent agents such as chatbots or LLMs are often unclear to end users. This lack of transparency forces users to guess whether the agent is acting in the best interest of the user's goals or that of the company or organization that designed it. In this paper, the authors introduce the concept of trust as a way to simplify such interactions.

Concretely, the evolutionary viability of trust-based strategies is explored in repeated games using concepts from evolutionary game theory. To do so, the authors introduce two new trust-based strategies, namely TUC and TUD, which start off as a standard reciprocal strategy such as Tit-for-Tat until they reach a predefined trust-threshold. Once this threshold is reached, both strategies will start trusting their opponent, in which case TUC will stop verifying its opponent's action each turn and TUD will start to defect unconditionally.

This mechanism is crucial to lower the opportunity cost which consists of verifying and keeping track whether an opponent is actually cooperative. When opportunity costs are taken into account, trust-based strategies outperform traditional strategies like Tit-for-Tat in a number of situations. In particular, the authors show that this is the case when the opportunity cost is sufficiently large and each interaction lasts for a minimum number of rounds. Moreover, the paper highlights that the presence of trust-based strategies increases the overall frequency of cooperation in the population. From these results the authors conclude that trust fits well in situations where the interactions are repeated over many rounds and the result is of intermediate importance.

Reproducing the experiments described in the original paper yielded identical results to the ones reported by the authors. This lead to the same conclusions as in the original paper, as well as some new insights into the different experiments. Finally, we also considered some modifications to the experiments as well as extensions to the original work.

1 Introduction

The paper by T.A. Han et al. [1] which we reproduced considers the concept of trust between agents and humans as a mechanism to alleviate the cost of having to verify each turn the opponents action. More specifically, the authors introduce the opportunity cost in repeated games as a means to incorporate the cognitive cost necessary to memorize past actions as well as the cost to verify the action taken by the opponent. This cost can often be overlooked or

underestimated, but can in certain scenarios be of crucial importance.

An example described in the paper is the situation where a user interacts with an artificial intelligence agent, such as a virtual sales agent or recommender system. In such a situation it is never truly clear to the end user whether the agent is cooperating by helping the user find what it is looking for, or defecting by serving a hidden goal programmed in its system such as recommending a pricier item. This lack of transparency brings risk to the conversation, which requires trust. A particularly relevant example of this, which was not mentioned in the paper, but which we found very fitting in this research, is the interaction between a user and a Large Language Model (LLM). This situation is very relevant these days and fits this research due to amount of hallucinations that they are known for. In order to examine the evolutionary viability of trust-based strategies in these situations, the authors introduce two new conditional strategies named TUC and TUD. They behave identical to Tit-For-Tat until a certain level of trust is reached. Once this level is reached, TUC will no longer verify its opponent's action each round while TUD will defect unconditionally no matter what happens next. In order to prevent itself from being exploited, TUC will still verify its co-player's action with a probability p to save on opportunity costs. If its opponent defects and TUC verifies that round, TUC will switch strategies and continue as TFT until the end.

The main contributions of the authors are the introduction of trust as a cost-saving mechanism in repeated games, discovering out in which situations trust is beneficial, and finally investigate the trade-offs of trust in repeated interactions.

The main results of this research show that trust outperforms classic reciprocal strategies when the opportunity cost is non-zero. Another important result observed by the authors is that trust increases overall cooperation in a population. In combination to this, they conclude that trust can only be successful in games that are repeated often and where the importance of the game is of *intermediate* importance.

2 Related Work

Looking through the literature we found three interesting papers that are closely related to the original one.

1. The evolution of trust as a cognitive shortcut in repeated interactions [2]

(i) This study uses the exact same model as the work reproduced in this report. It defines trust as a shortcut by which agents stop paying to verify their partner’s moves. Through simulation, the authors try to find out whether trust-based strategies can successfully compete against other strategies like Tit-for-Tat in an evolving population.

(ii) **Difference:** The paper we are reproducing focuses purely on the *Prisoner’s Dilemma*. This paper extends the analysis to include other social dilemmas, more specifically *Stag Hunt* and *Snowdrift*. By doing so, the authors show that the success of trust-based strategies depends heavily on the specific risks and rewards of the game being played. In addition to this, the paper introduces noise to a player’s own actions to find out how resilient trust is against (unintentional) mistakes.

2. A folk theorem in infinitely repeated prisoner’s dilemma with small observation cost [3]

(i) This paper addresses the exact same constraint found in our report: playing the *Prisoner’s Dilemma* in an environment where observing your opponent’s action has a cost. The goal is to mathematically prove that cooperation can still exist even when information is not free.

(ii) **Difference:** The main difference is the theoretical method used. Our report relies on *Evolutionary Game Theory* to simulate how new strategies survive in a population. This paper uses *Standard Game Theory*, focusing on mathematical proofs to find what agent should do, instead of simulating what evolving agents actually do.

3. Exploration dynamics in evolutionary games [4]

(i) A significant portion of papers on evolutionary game theory assume that mutations are rare. This paper suggests that this assumption could be misleading. Although the paper doesn’t explore trust-based strategies, it shows that mutation rates could affect the evolutionary outcomes.

(ii) **Difference:** The main difference is that our report focuses on trust and monitoring costs under rare mutation whereas [4] shows our frequently evolutionary dynamics in social dilemmas changes when the mutations are not rare. It does not consider trust in repeated interactions. Additionally, it focuses on generic strategies.

3 Practical aspects of the reproduction

In order to reproduce the experiments performed on the trust-based strategies discussed in the paper, we had to use the Small Mutation Limit (SML) to simulate the evolutionary dynamics of the five different strategies in a finite population. But first we had to calculate the payoff matrix to obtain the transition probabilities required to calculate the stationary distribution between the different strategies.

We did not find any existing implementations online or pseudo code in the paper itself, so resorted to implementing everything ourselves using python and Jupyter Notebooks. We implemented this using one function which takes the different

	AllC	AllD	TFT	TUC	TUD
AllC	R	S	R	R	$\frac{\theta R + (r-\theta)S}{r}$
AllD	T	P	$\frac{T + (r-1)P}{r}$	$\frac{T + (r-1)P}{r}$	$\frac{T + (r-1)P}{r}$
TFT	$R - \epsilon$	$\frac{S + (r-1)P}{r} - \epsilon$	$R - \epsilon$	$R - \epsilon$	$\frac{\theta R + S + (r-\theta-1)P}{r} - \epsilon$
TUC	$R - \frac{\theta\epsilon}{r} - \frac{p(r-\theta)\epsilon}{r}$	$\frac{S + (r-1)P}{r} - \epsilon$	$R - \frac{\theta\epsilon}{r} - \frac{p(r-\theta)\epsilon}{r}$	$R - \frac{\theta\epsilon}{r} - \frac{p(r-\theta)\epsilon}{r}$	$\frac{\theta R + S + (r-\theta-1)P}{r}$
TUD	$\frac{\theta R + (r-\theta)T - \theta\epsilon}{r}$	$\frac{S + (r-1)P}{r} - \epsilon$	$\frac{\theta R + T + (r-\theta-1)P - \theta\epsilon}{r}$	$\frac{\theta R + T + (r-\theta-1)P - \theta\epsilon}{r}$	$\frac{\theta R + (r-\theta)P - \theta\epsilon}{r}$

Figure 1: Payoff matrix (source: [1])

variables as function parameters in order to easily test various social dilemmas.

Our payoff matrix implements the formulas shown in (2) of the original paper [1]. The matrix contains some interesting fitness scores due to the fact that it takes the cost of monitoring the action of the co-player into account. As can be seen in figure 1, the fitness for fixed strategies does not change. Conditional strategies such as TFT, TUC and TUD have to pay an opportunity cost depending on how they choose to perform their action at that moment. An interesting example is the fitness score

$$R - \frac{\theta\epsilon}{r} - \frac{p(r-\theta)\epsilon}{r}$$

that TUC displays against AllC. It is calculated by subtracting two terms from the original reward R . The first term represents the average opportunity cost per round paid during the initial trust-building phase where TUC has to verify the co-player’s action every round for θ rounds, while the second represents the phase in which it trusts the AllC player and verifies the co-player action with probability p , paying opportunity cost ϵ each time. This example is rather straightforward due to the fact that AllC will always cooperate, leading to a simple first and second term.

Once the payoff matrix was implemented, we used the `eggttools` library to create a game using `Matrix2PlayerGameHolder`, then calculated the stationary distribution of the population using the pairwise comparison. The reason why we used this package is that it directly implements the pairwise comparison using the Fermi equation, which is the exact same evolutionary dynamic model described in the paper as well as the method that the authors used themselves.

4 Methods

As mentioned earlier, the authors of the paper resort to using methods for finite populations to simulate the dynamics between the *trust-based strategies* and the *classic strategies*. We used `eggttools` to calculate the stationary distribution of the different populations, as well as some other standard libraries such as `matplotlib` and `numpy`. In addition to this, we used Overleaf’s LLM assistance/AI Assist mode to help rewrite a unclear sentences. This usage was quite limited as the tool had a low limit for non-paying users and did not help significantly with the writing of this report.

LLMs were used solely to assist with data visualization techniques. No source code was passed to the model, nor was any AI-generated code included in the final implementation. The models were used exclusively to identify necessary library functions to achieve specific visual results. An example of such a query used is: ‘I want to

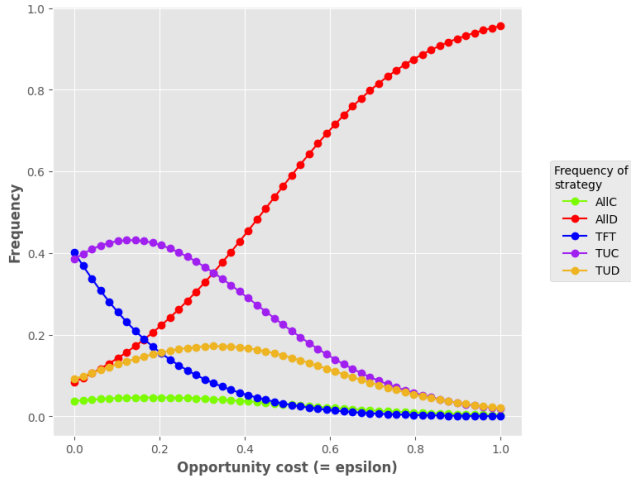


Figure 2: Frequency of strategies in function of opportunity cost ϵ

fill the area between two curves with a color, which methods can be used to do so?

Finally, we also used Google Gemini (Thinking, free version) to help summarize a couple of related papers. Based on the findings mentioned in the summaries, we went through the paper to read the original paragraph and verify its correctness. We found this tool quite effective at creating coherent summaries that match the content of the papers themselves without containing too many hallucinations.

5 Results

The first experiment we reproduced was the one in which the authors investigate the impact of the opportunity cost on the different strategies and by extension the overall cooperation on the overall population. As can be seen in figure 2, the proportion of each strategy is very much dependent on the opportunity cost.

Our results are almost identical to the original paper, and highlight that TUC is the most successful strategy when the opportunity cost is relatively low. This trend quickly changes once the opportunity cost rises above approximately 0.3, after which AllD becomes the dominant strategy.

This dramatic change is due to the fact that once the opportunity cost reaches that point, verifying an opponent becomes too expensive and defecting becomes more profitable. The experiment also highlights the advantage that TUC has over TFT when ϵ is non-zero. This is thanks to the trust mechanism incorporated into TUC that allows it to avoid paying the opportunity cost with a probability $1 - p$ as soon as it has reached the trust threshold θ .

A final interesting observation from this figure, is that when the opportunity cost is nil, both TUC and TFT are the most common strategies. This phenomenon can be attributed to the fact that both strategies can cooperate between themselves while avoiding exploitation by paying a non-existent opportunity cost.

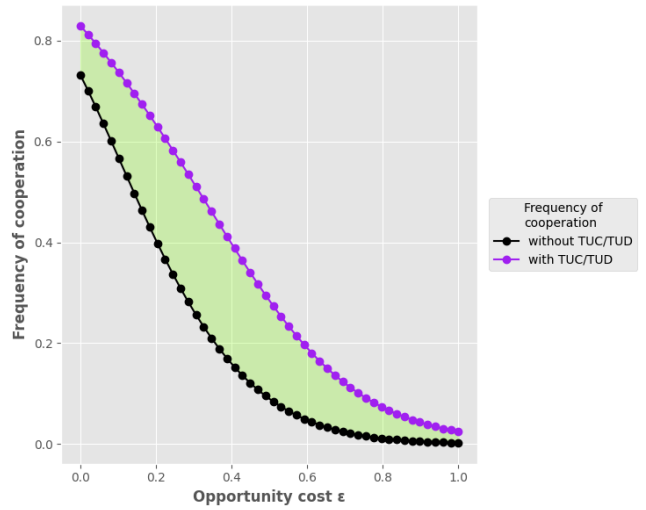


Figure 3: Frequency of cooperation in absence or presence of trust-based strategies TUC and TUD, as a function of the opportunity cost ϵ

5.1 Trust as a mechanism to reduce opportunity costs

Figure 3 shows the frequency of cooperation as a function of the opportunity cost ϵ . We considered two cases: a baseline population without trust-based strategies (AllC, AllD and TFT) and an extended setting in which trust-based strategies were also included (TUC and TUD). In both situations we observed the same trend: as ϵ increases, the frequency of cooperation decreases. This observation seems to confirm that an expensive monitoring costs decreases the evolutionary viability of cooperative behaviour within the population.

However, regardless of the value of ϵ , there is more cooperation in populations when trust-based strategies are present. The difference in cooperation with or without trust-based strategies starts small for low values ϵ , but becomes more pronounced at the intermediate values of ϵ , and reduces again as ϵ approaches 1. For low values of ϵ , cooperation is already prevalent due to the fact that TFT performs strongly in this situation. As ϵ increases, the presence of TFT declines and cooperation becomes more sensitive to the presence of trust-based strategies.

As a result, cooperation is consistently higher when trust-based strategies are present, with the largest difference observed at intermediate values of ϵ . To conclude, the reproduced results show that allowing trust-based strategies leads to a higher stationary level of cooperative behaviours across the entire range of opportunity costs which is consistent with the findings of the original paper.

5.2 Importance of the game

Figures 4 and 5 visualize how trust, and by extension cooperation, depends on the importance of the game (expressed with γ) as well as the expected number of interaction rounds r . The importance of the game determines how valuable cooperation and defection

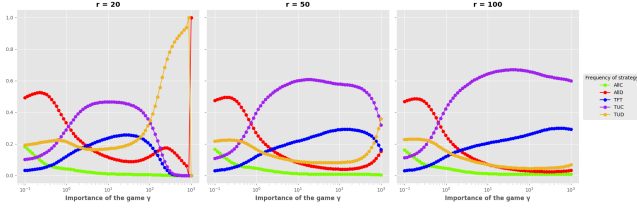


Figure 4: Frequency of strategies as a function of the number of rounds r and importance of the game γ (logarithmic scale)

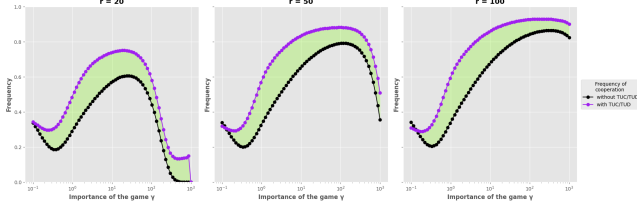


Figure 5: Frequency of cooperation in absence or presence of trust-based strategies TUC and TUD, as a function of the number of rounds r and importance of the game γ (logarithmic scale)

are relative to the opportunity cost, while the number of rounds affects the time available for trust to be built.

Figure 4 shows how the frequency of the different strategies varies with regards to γ when the rounds lasts for 20, 50 or 100 interactions. One of the first things that can be observed, is that defection dominates when γ is low no matter the interaction length. This trend can be attributed to how γ modifies the rewards of the game compared to the opportunity cost ϵ . To be more clear, when the importance of the game is negligible, the opportunity cost becomes far too expensive compared to the rewards of the game. This makes it more tempting to defect since it is not viable to monitor an opponent's action.

Inversely, when γ is high the importance of the game changes completely. Short interactions are now dominated by defection as there is a high temptation to defect and insufficient time to develop trust between agents. A completely different trend can be observed with longer interactions. In this situation, TUC dominates all other strategies because it is able to cooperate with the cooperative strategies AllC, TUC and itself while preventing exploitation from defective strategies. This is likely due to the fact that TUC is rather untrusting in this experiment, which will be discussed in section ??.

Figure 5 shows the frequency of cooperation inside a population with or without trust-based strategies. For all interaction lengths, it can be concluded that cooperation is higher when trust-based strategies are present. Cooperation is most successful when interactions last longer and the game is important. For shorter interactions, cooperation eventually declines at higher values of γ , especially in the absence of trust-based strategies. This is to be expected, as

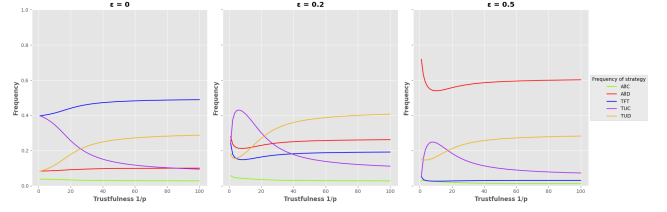


Figure 6: Frequency of strategies as a function of the opportunity cost and trustfulness $1/p$

higher values of γ increase the temptation to defect. The shape of the cooperation curves closely follows the frequency of TUC, suggesting that increased cooperation is primarily driven by the prevalence of trust-based cooperation. A notable exception occurs for short interactions and high γ , where cooperation is still higher in the presence of trust-based strategies despite the absence of TUC. In this case, the increase in cooperation is due to TUD cooperating for a fixed number of rounds, whereas AllD never cooperates.

5.3 The impact of the trustfulness parameter

Figure 6 investigates the impact of the trustfulness parameter ($1/p$) on the stability of cooperation. The variable $1/p$ represents the average number of rounds an agent waits before re-verifying a partner.

The frequency of cooperation of the trust-based strategy behaves unexpectedly as the trustfulness ratio $1/p$ increases. When trustfulness is low, TUC checks its opponent often and therefore has to pay the opportunity cost oftentimes. As a result, it is less competitive and remains relatively unsuccessful compared to other strategies. As trustfulness increases, TUC checks its opponent less often and leads to a reduction in its opportunity cost resulting in its frequency in the population increasing. This increase continues gradually until an intermediate value of trustfulness is reached at which point TUC becomes the most frequent strategy.

When trustfulness increases further, TUC becomes too trusting and checks its partner rarely. Once this stage is reached, TUC is vulnerable to exploitation by the trust-based defector TUD, which defects after trust has been established and exploits TUC for a certain number of rounds each time. This leads to a decrease in the frequency of TUC when trustfulness is high. The strategy that takes over in this situation depends on the value of the opportunity cost ϵ . For low values of ϵ , TFT becomes dominant due to the low opportunity cost. For intermediate ϵ , TUD dominates due to its ability to exploit trust-based strategies efficiently. Finally, for high values of ϵ , AllD dominates because both monitoring and cooperation are too costly. Overall, these results demonstrate that TUC is most successful at an intermediate level of trustfulness concluding that TUC works best when it is neither too careful nor too trusting.

Figure 7 visualizes the frequency of cooperation inside a population both with and without trust-based strategies for various values of opportunity cost ϵ . Similarly to figure 6, trustfulness represents how often a trust-based strategy checks its opponent's behaviour: low $1/p$ values corresponds to frequent checking, while high $1/p$

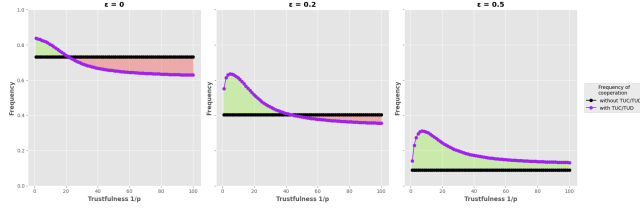


Figure 7: Frequency of cooperation in absence or presence of trust-based strategies TUC and TUD, as a function of the opportunity cost and trustfulness $1/p$ (average number of rounds between checking event).

corresponds to very infrequent or rare checking. For moderate and high values of ϵ , the presence of trust-based strategies leads to a clear increase in cooperative behaviour regardless of the trustfulness. When trustfulness is low, cooperation levels are similar with and without trust-based strategies since checking frequently makes trust-based behavior similar to TFT. However, unlike previous experiments, figure 6 also shows that trust-based strategies can reduce cooperation under certain conditions. When the opportunity cost ϵ is low and trustfulness is high, cooperation becomes lower in the presence of trust-based strategies than in their absence. In this situation, overly trusting TUC players become exploited by TUD, which increases the frequency of defective behavior and reduces the overall level of cooperation. In contrast, when trust-based strategies are absent, TFT avoids this outcome by constantly checking to prevent exploitation on the long-term. To conclude, these results show that trust-based strategies can positively impact cooperation around intermediate levels of trustfulness. Too much trust enables exploitation while too little trust makes the total opportunity cost of constant verification too expensive.

5.4 The cost of verification

Figure 8 visualizes how the opportunity cost (ϵ) impacts the frequency of the different strategies similarly to figures 2 and 3. However, the experiment is now ran with different trust thresholds to highlight the impact that this variable has on trust and by extension cooperation.

Impact of the Trust Threshold θ : The trust threshold θ defines how many rounds the trust-based agent will *constantly* verify a partner's actions before deciding whether to *occasionally* verify them (dictated by the parameter p). Comparing the top row ($\theta = 5$) and bottom row ($\theta = 10$) reveals important details. In the top plot ($\theta = 5$), TUC survives until ϵ is around 0.7. In the bottom plot ($\theta = 10$), TUC crashes earlier, around $\epsilon = 0.5-0.6$. The reason is that by increasing the Trust Threshold θ , TUC suffers because it has to pay the verification cost for every single round until the threshold is reached. But overall the results are very similar, highlighting how **robust** trust-based strategies are in various settings.

Comparison to Original Results: Our results mirror those of the original paper. We see the same exact decline in cooperation when costs rise. The point where AllD takes over TUC also matches what was given in the original paper.

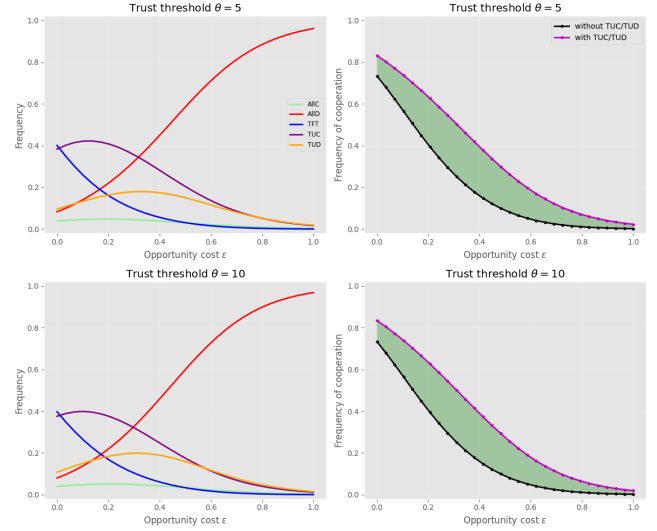


Figure 8: Frequency of strategies and their cooperation frequency in function of opportunity cost ϵ

5.5 The impact of interaction length

Figure 9 displays the behavior of the system as the *importance* of the game (γ) increases logarithmically from 10^{-1} to 10^3 . This parameter scales the payoff rewards to the fixed cost of verification. At the same time, this experiment is conducted over different interaction lengths to showcase the robustness of trust when interaction length varies.

Short games ($r=20$): In short repeated games, trust-based strategies have a harder time building up a cooperative behaviour with their peers. At low importance, the population is mixed. However, a transition occurs at high important ($\gamma > 10^2$). The TUD strategy (orange line), which verifies partners only to start exploiting them, suddenly jumps up. This causes cooperation to drop quickly (bottom plot). **Important note:** In the reproduction, specifically for the short game ($r=20$), a **sharp rise/drop** is visible at the far right of the plot where selection intensity is highest. Up until that point, the figure fully mirrors the one in the original paper that it is reproducing. The reason is that some of the entries in the transition matrix are close to 1 in that point, which results in more than one eigenvalue of magnitude 1 (according to the warning in Python). This means the Markov Chain is degenerate. This problem is not easily fixable, because Numpy does not support **128-bit floating point** values, which means we are stuck with the **64-bit floating point** values that are causing the **precision limitations**.

Long games ($r=50, 100$): In longer interactions, the cooperative trust-based strategy TUC (purple) becomes dominant. The longer duration allows TUC agents to find and exclude defectors more effectively. As a result, the bottom plots for $r=50$ and $r=100$ show a strong green region, which tells us that cooperation stays high.

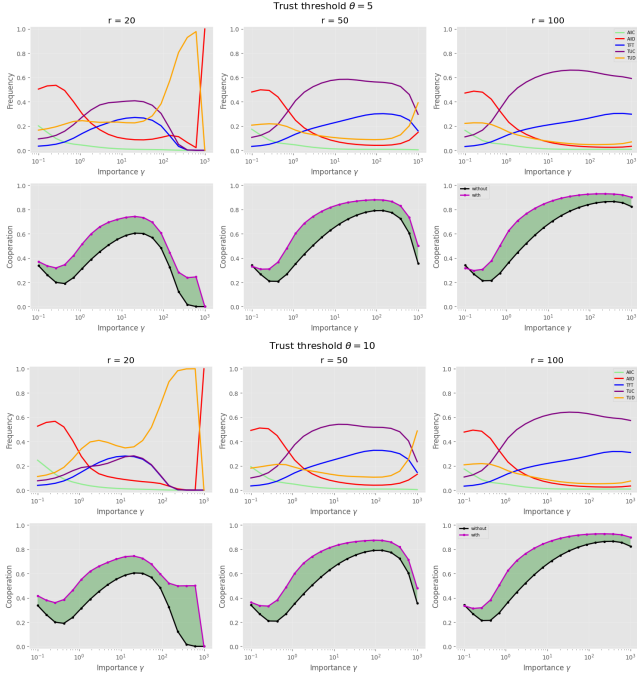


Figure 9: Top: Frequency of strategies as a function of the number of rounds r and importance of the game γ (logarithmic scale). Bottom: Frequency of cooperation in absence or presence of trust-based strategies TUC and TUD, as a function of the number of rounds r and importance of the game γ (logarithmic scale)

5.6 Trustfulness under different thresholds

Repeating the experiment on the trustfulness of TUC and TUD with trust thresholds 5 and 10 leads to the results visualized in figure 10. This experiment demonstrates how robust trust can be when the threshold is modified significantly.

Impact of the trust threshold $\theta=5$ vs $\theta=10$:

Short games ($r=20$): If $\theta = 10$, TUC agents spend 50% of the game (10 out of 20 rounds) testing before determining whether to trust or not its partner. This hinders TUC from profiting from the reduction in opportunity costs, as the remaining ten rounds are insufficient to offset the heavy cost of the initial trust-building phase. This is an important reason why **TUC is weaker** and AllD/TUD is stronger in short games where θ is high.

Long games ($r=100$): If $\theta = 10$, trust-based strategies only spend 10% of the game testing their opponent before potentially becoming trustful. The difference between testing for 5 rounds vs 10 rounds is negligible in this case because there still remain 90 rounds where the agents can enjoy the benefits of less verification. It is worth noting that this research does not take noise into account. Would this have been the case, then the results could be significantly altered. The reason for this is that lower values of θ can render a strategy more forgiving in situations where an agent defects by mistake for example. This important aspect is also mentioned in our discussion as an extension of the original paper.

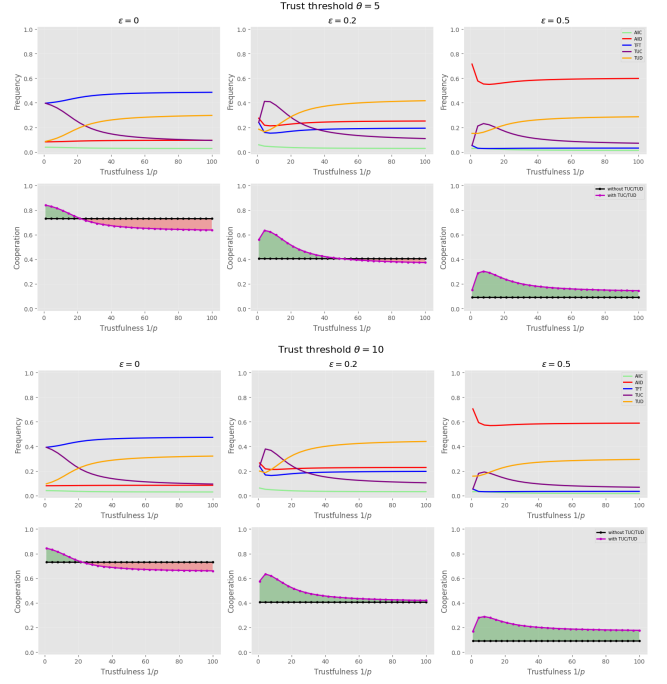


Figure 10: Top plots: Frequency of strategies as a function of the opportunity cost and trustfulness $1/p$ (average number of rounds between checking event). Bottom plots: Frequency of cooperation in absence or presence of trust-based strategies TUC and TUD, as a function of the opportunity cost and trustfulness $1/p$

Comparison to original results: Our simulation successfully reproduces the exact results from the original paper, **except** for the $r=20$ case in the right hand side of the plot, for reasons that we have outlined. Just like in the paper, TUC is the dominant trust strategy when the games are long.

The impact of the trust threshold θ : Comparing $\theta = 5$ and $\theta = 10$, the cooperation advantage (green area) is generally equal or even slightly larger in the $\theta = 10$ case. Since TUD agents must mimic cooperation during the initial period to survive, increasing the threshold to $\theta = 10$ forces these impostors to cooperate for twice as many rounds (10 vs 5) before they can begin defecting. This extended period of forced cooperation contributes to a higher aggregate cooperation level.

6 Discussion and Future Work

6.1 Interpretation of Results

Our reproduction confirms the main finding of the original paper: trust can act as a shortcut that enables cooperation in environments where the act of observing and keeping track of a partner's action are not free. We successfully showed that the *Trust-based Cooperator* (TUC) strategy performs better than standard Tit-for-Tat when there is noise in the system, as long as the cost of verification is not too high. Finally, we must address the numerical instability

observed in the short game ($r = 20$) reproduction in Figure 7. At the far right side of the plots, where the selection intensity is highest, the results display a sharp, unnatural divergence. This is a technical artifact caused by the Markov Chain transition matrix becoming degenerate. As the selection intensity increases, the transition probabilities approach 0 or 1, causing floating-point precision errors that the standard solver cannot handle. This leads to the weird behavior visible at the very end of the plots, which is a calculation error rather than a genuine result.

The different experiments presented by the authors highlight that using trust successfully comes down to balancing the risk of exploitation with cheaper cooperation. Understanding this balance is crucial to manage the trade-off between both.

Although very different in execution, a similar paper that tackles the problem of cooperation and by extension trust in PD, is the Co-evolution of pre-play signaling and cooperation [5]. This is the same paper that we were tasked to implement in one of the assignments. The difference between both is that [5] tackles one-shot games where pre-signaling is used as a way to trust or discriminate between strategies.

6.2 Future Analysis and Extensions

Based on our findings, we propose three specific directions for future work to advance the results of the original paper.

1. Evolving the Trust Threshold: Currently, the trust threshold θ is a fixed number chosen by the researchers (e.g., 5 or 10). In a real setting, this threshold would likely adapt to the environment. A future analysis should implement θ as a variable that can evolve. Allowing θ to evolve with the strategy would provide a more realistic picture of how trust emerges.

2. Imperfect Verification (noise): The current model assumes that if an agent pays the cost, they get perfect information about their partner. Real-world verification is rarely 100% accurate. Future work should introduce "noisy verification," where paying the cost still leaves a small chance of misinterpreting the partner. This would test if the TUC strategy remains robust when checking is expensive and unreliable.

3. Extend Comparison: An interesting extension to this research would be to extend the comparison between trust-based strategies with strategies win-stay-lose-shift, grim trigger, generous TFT and others. This extension should naturally be paired with the second proposal to investigate how well trust performs as a forgiving strategy that enables cooperation in difficult dilemma's such as PD.

4. Technical Robustness: Finally, our difficulties with the numerical stability of the Markov Chain suggest that standard Python tools (using 64-bit precision) are insufficient for analyzing selection in these specific game theory models. Future implementations should use higher-precision environments (like 128-bit floating point libraries) to prevent the artificial "crashes" we observed, ensuring that the results at the extremes remain accurate.

7 Conclusions

In this project, we reproduced the results of the paper "When to (or not to) trust intelligent machines: Insights from an evolutionary game theory analysis of trust in repeated games" as best as we could. The aim was to apply the model and verify whether the main results of the original paper could be obtained again by carefully following the described model. Overall, our reproduced results closely match those reported in the paper confirming the reliability of the original analysis.

Through our reproduced results, we observed the same key pattern as highlighted by the authors. The reproduced figures confirm that trust-based strategies perform quite well in promoting cooperation in repeated PD games when the cost of constantly monitoring the other player is non-negligible. Compared to traditional strategies such as Tit-for-Tat, TUC can maintain high levels of cooperation while reducing the monitoring opportunity costs, especially when it comes to long term interactions. Additionally, we confirmed that trust is most effective in interactions of intermediate importance and adding to this point, excessive trust can make players vulnerable to exploitation by trust-based defectors such as TUD. These results were consistent with the results of the original paper.

From the experience perspective, the paper has provided us with a deeper understanding of the different mechanisms possible to promote cooperation in difficult social dilemma's such as the Prisoner's Dilemma. It was quite useful but also challenging. The model was described clearly in the original paper which made the reproduction somewhat easier, however, there was no algorithm or pseudo code mentioned in the original paper. A somewhat annoying fact is the problem we encountered regarding the numerical instability when computing figure 9, but overall everything we are pleased with the end result.

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